

How Do We Actually Govern Indoor Air Quality?

Technical Governance Reference | Eight-Anchor Examination | School-Air HOLD Scenario

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A practitioner-facing explanation of how environmental observation can be preserved as evidence, examined against applicable authority and established standing, and connected to a bounded, verified consequence.

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Status: Educational reference only. The worked school scenario, readings, instrument identities, decision records and illustrations are hypothetical. No baseline has been frozen; no formal disposition or live building authorization is asserted. No EPA, ASHRAE, school district or other institutional affiliation or endorsement is claimed.

How Do We Actually Govern Indoor Air Quality?

From environmental reality to verified outcome, using **admissible evidence** and **authorized execution**.



THE GOVERNING QUESTION
Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality **NOW?**



WORKED EXAMPLE · A SCHOOL NEEDS VENTILATION. THEN THE OUTDOOR AIR CHANGES.

An occupied classroom's CO₂ level supports consideration of additional ventilation. Before a prepared HVAC command executes, outdoor PM2.5 rises with wildfire smoke. The earlier observation does not automatically authorize the same command under changed conditions.



1 | Scope and governing distinction

Monitoring observes environmental conditions. A sensor, dashboard, model, recommendation or preprogrammed command does not by itself establish permission to change a building. TA-14 examines the boundary between a proposed consequence and authorized execution.

Atmospheric Integrity Records (AIR) address preservation, attribution and continuity of environmental observations. Environmental Integrity Governance (EIG) concerns responsible examination of environmental evidence and its relationship to decisions. These functions complement - but do not replace - qualified engineering judgment, law, building policy, commissioning or equipment safeguards.

2 | Eight-anchor chain

01 REALITY	Identify the actual occupied-space condition and operating context.
02 RECORD	Preserve raw readings, timestamp, location, instrument identity, calibration context and HVAC state.
03 CONTINUITY	Preserve attribution across custody, storage, transformation, gaps and changing conditions.
04 ADMISSIBILITY	Examine whether the evidence is current, relevant, reliable and sufficient for this proposed action.
05 BINDING	Connect the proposal to evidence, policy, equipment, scope, responsible parties and constraints.
06 COMMIT	Confirm applicable authority and established standing at the actual decision point.
07 EXECUTION	Apply ALLOW, HOLD, DENY or ESCALATE before a bounded command crosses into physical action.
08 OUTCOME	Verify and preserve what happened - including a prevented action - as new reality.

3 | Worked school-air scenario

A hypothetical occupied classroom has elevated CO₂, leading a responsible party to consider increasing outdoor-air ventilation. Before the prepared command is executed, outdoor PM_{2.5} rises during a wildfire-smoke event. The original record remains preserved, but the conditions supporting the proposed intervention have materially changed.

Illustrative inputs, not measured or verified data

- Original indoor CO₂: 1,480 ppm; illustrative outdoor PM_{2.5}: 8 micrograms/m³.
- Proposed action: increase AHU-1 outdoor-air fraction from 35% to 50% for Classroom 4B.
- Changed outdoor PM_{2.5}: 85 micrograms/m³ before execution.
- Illustrative disposition: HOLD the previously prepared command pending re-examination.

The HOLD is not a blanket instruction to reduce ventilation or disregard indoor-air risks. It prevents this particular proposed command from being released on stale evidence. Qualified local decision-makers must consider current measurements, occupancy, filtration, equipment capability, relevant health and safety requirements, and available alternatives.

4 | Disposition and local execution boundary

ALLOW - Current admissible evidence, applicable authority and established standing support the bounded action.

HOLD - The proposal is not released while a correctable evidence, timing, policy or authority condition remains unresolved.

DENY - The proposal is not permitted under applicable conditions or authority.

ESCALATE - The matter requires additional evidence, higher or different authority, or a defined review path.

Authority context may be transmitted across systems or organizations, but execution authority must be established locally for the specific equipment, action, timing and conditions. Technical capability is not execution permission.

5 | What must be preserved

Minimum record fields for a defensible examination include raw observation and units; measurement method and instrument identity; timestamp and time zone; location and occupancy context; calibration/quality limitations; HVAC state; outdoor-air context; record lineage and custodians; proposed command and bounded scope; policy and authority references; identity and standing of decision-maker; disposition with rationale; execution gate result; and observed outcome with any uncertainty.

01 | REALITY

Observed classroom and outdoor conditions, including the equipment context, must be distinguished from assumptions.

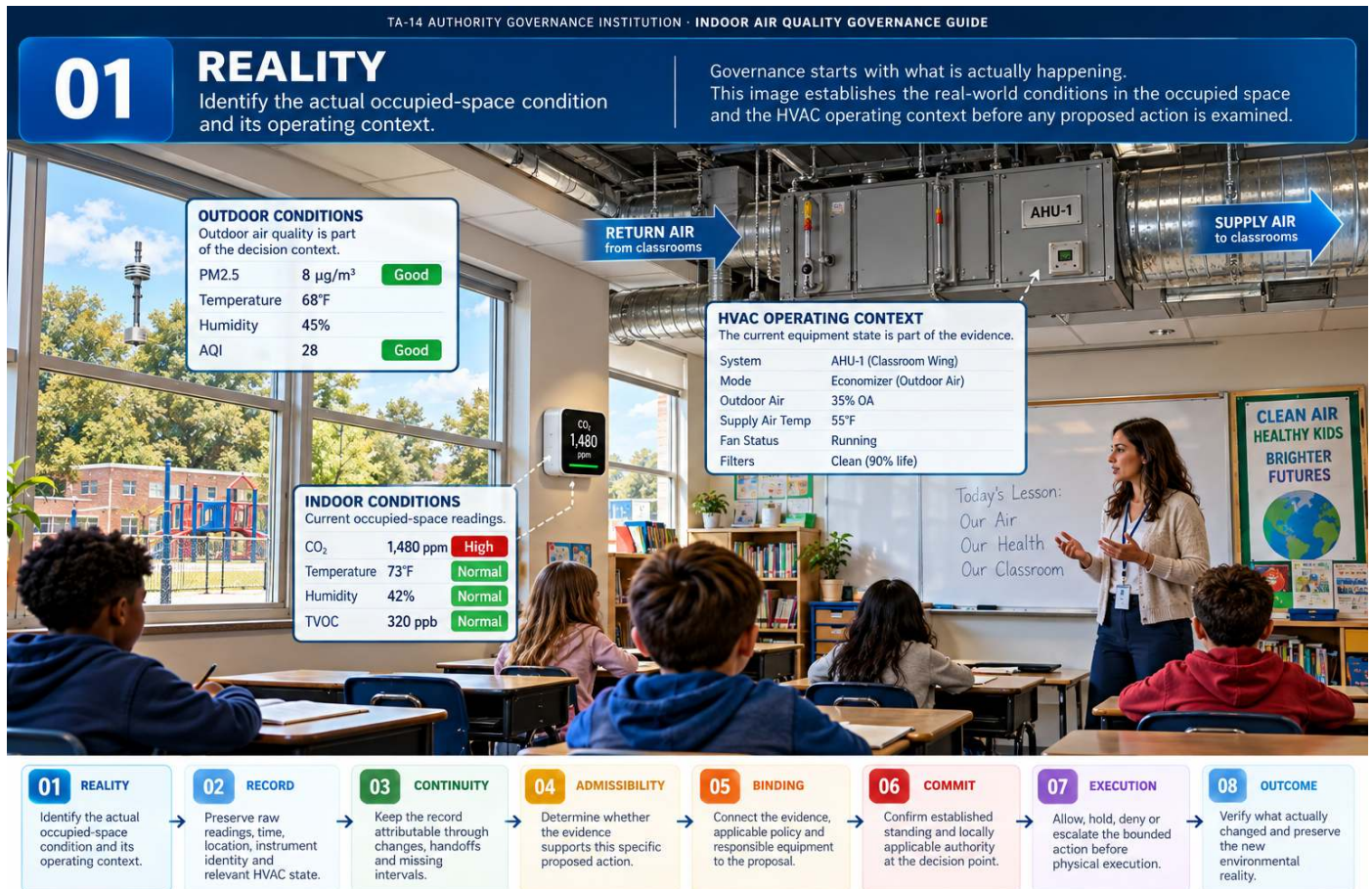


Illustration only. All depicted readings, identities, authorizations, checks and outcomes are hypothetical; a graphic is not an operational log or formal disposition.

02 | RECORD

The observation becomes a preserved, attributable record; the illustrative identifiers and readings are not live evidence.



03 | CONTINUITY

The original and subsequent environmental observations remain traceable through custody and changing conditions.



04 | ADMISSIBILITY

A valid CO2 observation may remain true while becoming insufficient to justify a particular ventilation command.

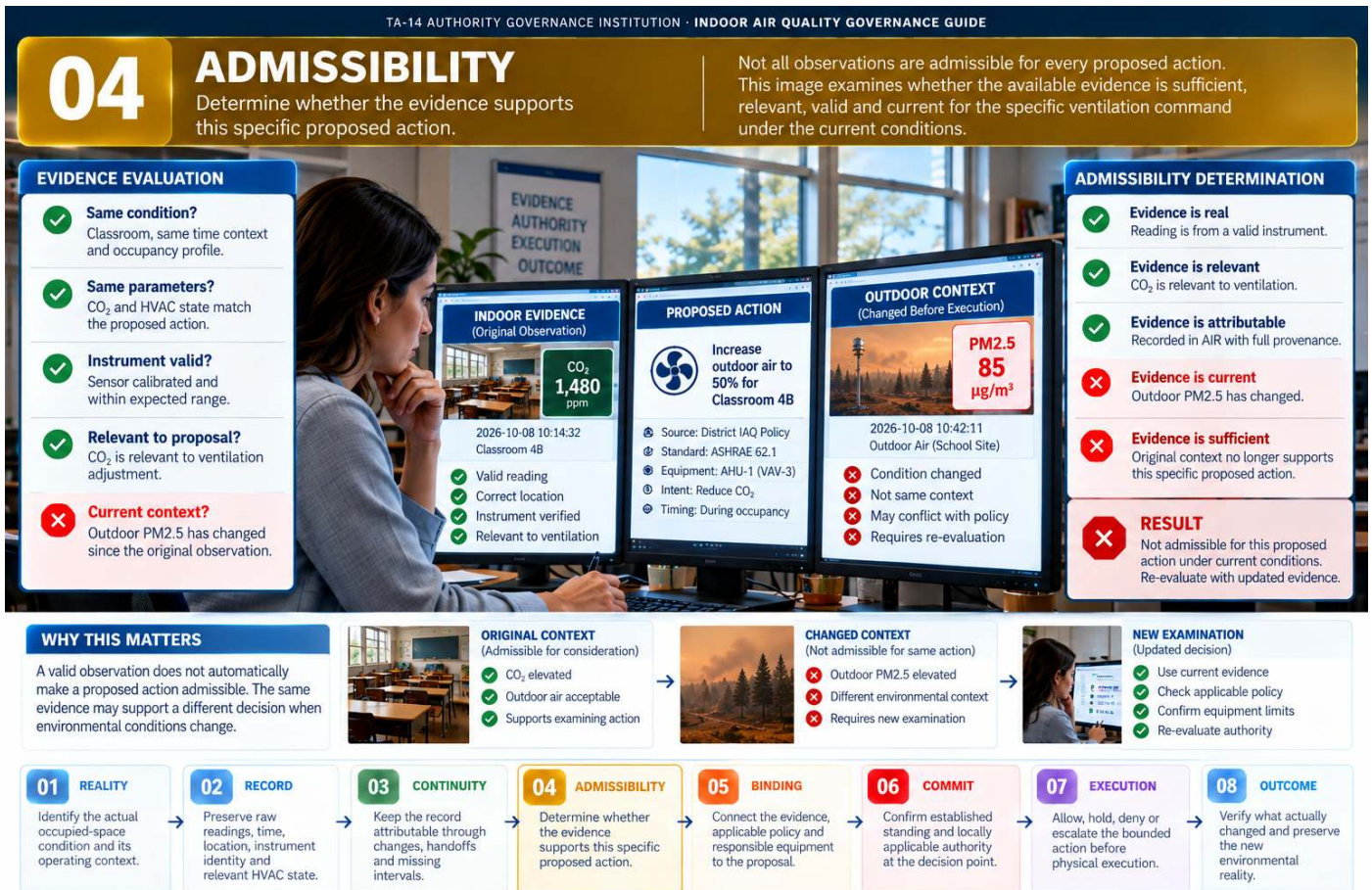
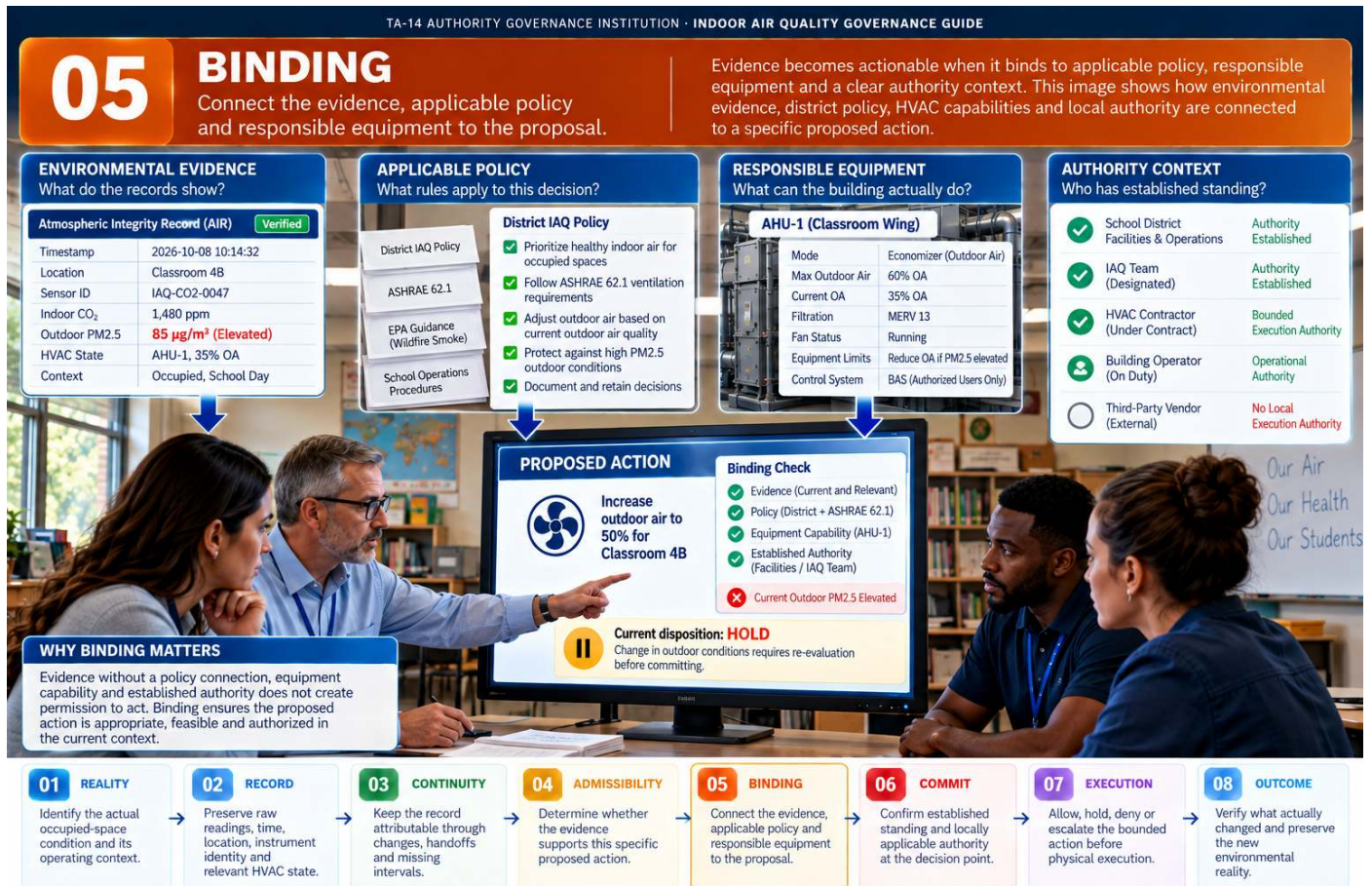


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05 | BINDING

Policy, equipment capability, responsible actors and current evidence must be linked to the same proposal.



01 REALITY

Identify the actual occupied-space condition and its operating context.

02 RECORD

Preserve raw readings, time, location, instrument identity and relevant HVAC state.

03 CONTINUITY

04 ADMISSIBILITY

05 BINDING

06 COMMIT

07 EXECUTION

08 OUTCOME

Illustration only. All depicted readings, identities, authorizations, checks and outcomes are hypothetical; a graphic is not an operational log or formal disposition.

06 | COMMIT

The illustrative decision is HOLD because changed context requires a new examination before authorization.



07 | EXECUTION

The proposed command is not released; the actual equipment state must be checked independently.



Illustration only. All depicted readings, identities, authorizations, checks and outcomes are hypothetical; a graphic is not an operational log or formal disposition.

08 | OUTCOME

Record the non-execution, verify observed building state and continue collecting evidence for the next decision.



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6 | Examination checklist

- Is the physical condition actually observed rather than inferred?
- Are original data and subsequent transformations preserved separately?
- Are source, location, time, instrument and relevant limitations attributable?
- Is the evidence still current for this specific proposed consequence?
- Have changed environmental conditions been recorded without overwriting the original?
- Are applicable requirements and equipment constraints identified?
- Is the decision-maker authorized and established as having standing here and now?
- Is the proposed action bounded by target, scope, duration and safety constraints?
- Was the disposition recorded before any execution request?
- Can the actual equipment response or non-execution be independently verified?

7 | Reference pathways

Indoor Air Governance Guide: <https://www.ta14exchange.com/indoor-air-governance>

Environmental Integrity Governance: <https://www.ta14exchange.com/environmental-integrity-governance>

Atmospheric and Environmental Records: <https://www.ta14exchange.com/environmental-records>

School-Air Governance Showroom: <https://www.ta14exchange.com/showrooms/school-air-quality-governance>

Public Technical Corpus: <https://www.ta14exchange.com/foundation/public-corpus>

Governance Registration: <https://www.ta14exchange.com/workspace/ai-governance/registry/register>

Reference integrity note

This publication is a TA-14 educational technical reference, not an adopted regulation, engineering specification, independent certification, completed field examination, or substitute for competent professional judgment. The linked materials are provided as navigation paths; publication and availability should be checked before operational use.

No admissible evidence. No admissible execution.